

Sportzfy

Requirement Analysis Report & Data Validation

This document details the formalized system requirements for the Sportzfy turf booking platform, validated against a localized market survey of 25 respondents from the Chittagong University of Engineering & Technology (CUET) campus community, alongside qualitative observations from local sporting networks.

I. Validated Core Requirements

The primary objective of transitioning users from manual coordination to a centralized digital marketplace is heavily supported by the user data.

Real-Time Availability Engine

Currently, 60% of surveyed players rely on manual phone calls for turf reservations. Social channels confirm frequent manual inquiries for open slots at specific times (e.g., 8 PM, 12 AM). The system must prioritize a live, real-time availability grid.

Targeted Search & Filtering

When selecting a turf, 40% prioritize timing/availability, while 32% prioritize price. Market analysis shows high demand for specific pitch formats (e.g., 7v7, 6v6) in particular areas. Search parameters must support location, sport type, and price filters.

Transparent Payments & Refunds

While 52% of users are open to paying upfront to lock a slot, a decisive 72% stated that an automated, transparent refund policy within the app is a crucial requirement. The payment gateway must support mobile banking integration (bKash, Nagad).

II. High-Priority Value-Add Features

These features scored exceptionally high during validation and address specific pain points within the community ecosystem.

- **In-App Split Payments:** A significant majority (76%) of respondents indicated that a split-payment feature would be "extremely useful" as booking costs are inherently shared among teams.
- **Matchmaking & Player Recruitment:** Market observation reveals a significant gap in team building, with groups frequently searching for opponents or specific roles (e.g., goalkeepers). Survey data confirms this, with 48% of users actively seeking other teams to play against. A dedicated "Find Opponents" or "Join a Match" module is highly validated.
- **1-Click Share Invites:** 76% of respondents deemed a 1-click share function "very important" for distributing turf locations and match timings to teammates via external messaging platforms.
- **Equipment Rental Integration:** 60% of players desire the capability to rent supplementary gear (footballs, bibs, gloves) directly through the application during the booking workflow.

III. Quality Control & Amenities

Ensuring facility quality and setting accurate user expectations are vital for platform trust.

- **Verified Reviews & Ratings:** Qualitative data highlights community friction regarding facility staff behavior. A verified post-booking review system is required to maintain accountability.
- **Facility Amenities Filter:** 48% of surveyed players consider clean washrooms and changing rooms a mandatory filter requirement, followed by drinking water availability (32%).

IV. Engagement & Retention Strategies

Features designed to drive repeat usage and enhance the user experience.

- **Loyalty & Rewards Program:** 60% of respondents are "very likely" to utilize the platform as their primary booking method if a loyalty points or discount structure is implemented.
- **AI Weather Alerts:** 60% of users expressed high interest in AI-driven rain alerts and weather forecasts specific to their booked time slots.
- **Communication Channels:** For booking confirmations and reminders, 52% prefer in-app push notifications, and 36% favor automated WhatsApp messages.

V. System Architecture Updates

Based on the validated requirements, the backend entity-relationship model requires expansion.

Entity	Proposed Expansions
Bookings	Add <code>split_payment_status</code> (boolean/string) and <code>equipment_rented</code> (JSON/array).
Turfs	Add <code>amenities</code> array (e.g., ["Washroom", "Water", "Parking"]) and <code>pitch_size</code> attributes (e.g., ["7v7", "6v6"]).
Matchmaking (New)	Create component to handle "Open Matches" where solo players can RSVP to incomplete rosters.